

Time-Step-Free Physics-Based Modeling of Retention Loss in Cryogenic 3-D NAND Flash

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Abstract—We propose a time-step-free physics-based model for retention loss in cryogenic 3-D NAND flash memory. By solving the first-order autonomous detrapping equation in closed form, the proposed model directly evaluates $\Delta V_{th}(t)$ from the initially programmed trapped charge distribution without transient time stepping. For a fixed number of targeted retention times, the computational cost is independent of the final physical retention time, enabling efficient short- and long-term retention prediction. The model is benchmarked against TCAD simulations calibrated only to the time scale of reported cryogenic retention data and configured to isolate detrapping loss by suppressing thermally activated transitions during retention. It reproduces the TCAD predicted retention transient and tunneling oxide thickness dependence for 3-5 nm while reducing the average runtime from >25000 s to <4 s.

Index Terms—time-step-free modeling, autonomous ODE, detrapping, 3-D NAND flash memory, retention loss modeling

I. INTRODUCTION

THE rapid increase in data storage demand, driven by emerging computing technologies such as artificial intelligence (AI) and high-performance computing (HPC), has raised significant concerns about the reliability of 3-D NAND flash memories, particularly regarding retention loss [1].

As storage capacity per cell increases, ensuring long-term stability of each bit's threshold voltage distribution over time scales from days to years becomes increasingly critical. Moreover, prior works have shown that trapped charge detrapping can still occur at cryogenic temperatures [2], [3], supporting models that treat detrapping as temperature independent [4].

In this study, a time-step-free physics-based model is developed for retention loss based on tunneling emission mechanisms in 3-D NAND flash memory. The proposed simulation framework uses the initially programmed trapped charge distribution as input and enables highly efficient computation by using closed-form solution of the autonomous detrapping equation, eliminating sequential transient time stepping.

II. MODELING

This section presents the modeling framework for time-step-free simulation of retention loss. First, a cylindrical Poisson solution is derived for the oxide–nitride–oxide (O/N/O)

gate stack, incorporating quantized multiple layers within the charge trap nitride (CTN) layer. After obtaining the detailed Poisson solution for these discretized layers, the threshold voltage shift (ΔV_{th}) is calculated using a determinant based method. Finally, based on previous studies of detrapping-induced emission rates [4], an autonomous equation governing retention loss behavior is proposed.

A. Detailed Poisson Solution

An electrical potential distribution obtained from the cylindrical Poisson equation for discretized layers in the ONO structure can be expressed as follows [4], [5].

$$\begin{cases} V_{Si}(r_{Si}) = V_{ch} \\ V_{TOX}(r) = V_{ch} + \frac{K_{TOX}}{\epsilon_{TOX}} \ln \frac{r}{r_{Si}} \\ V_{CTN,i}(r) = V_{CTN,i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{CTN}} \ln \frac{r}{r_{i-1}} \\ V_{BOX}(r) = V_{CTN,N}(r_N) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_N} \\ V_{BOX}(r_{BOX}) = V_G \end{cases} \quad (1)$$

where $i = 1-N$ denotes the index of each discretized slice of the CTN layer in the radial direction, and N is the total number of radial partitions.

Solving $\mathbf{x}$ vector can calculate the total $N + 2$ potential parameters.

$$\mathbf{x} = [K_{TOX} \quad K_1 \quad \cdots \quad K_N \quad K_{BOX}]^T \quad (2)$$

$$A\mathbf{x} = \mathbf{b} \quad (3)$$

Matrix A and vector $\mathbf{b}$ are constructed by enforcing a total of $N+1$ transverse electric field continuity conditions ($\epsilon_1 \mathcal{E}_1 = \epsilon_2 \mathcal{E}_2$) together with an additional boundary condition defined by reference potential.

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{TOX}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{BOX}} \end{bmatrix} \quad (4)$$

$$\mathbf{b} = [\frac{\Delta Q_0}{2\pi} \quad \frac{\Delta Q_1}{2\pi} \quad \cdots \quad \frac{\Delta Q_N}{2\pi} \quad \Delta V]^T \quad (5)$$

$$\Delta Q_i = \pi r_i^2 (n_{t,i} - n_{t,i-1}) \quad (6)$$

$$\Delta V = (V_G - V_{ch}) - \sum_{i=1}^N \frac{qn_{t,i}}{4\epsilon_{CTN}} (r_i^2 - r_{i-1}^2) \quad (7)$$

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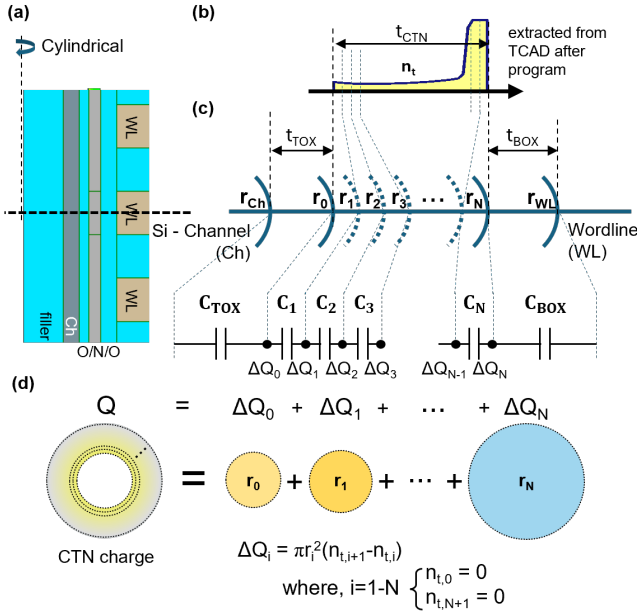


Fig. 1. (a) shows a radial slice of 3-D NAND flash memory, (b) shows TCAD-extracted post-program trapped charge distribution, (c) shows capacitance and node charge representation, and (d) shows the visualization of ΔQ_i for each node.

C_{TOX} , C_i , and C_{BOX} in (4) denote the capacitances of each cylindrical layer, as illustrated in Fig. 1(c). ΔQ_i in (5) represents the charge component associated with the cylindrical shell at radius r_i , which can be superpositioned as $Q = \sum \Delta Q_i$ as demonstrated on Fig. 1(d). Furthermore, (5) indicates that $b(\Delta Q_i, V_G - V_{ch})$ varies with the trapped charge distribution and the applied voltage conditions.

B. $Q - \Delta V_{th}$ Relation

For the ΔV_{th} calculation, the gate voltage difference required to maintain equal channel field conditions must be determined [4], [5].

$$F_{TOX}(r_{Si}) = \frac{K_{TOX}}{\epsilon_{TOX}} \frac{1}{r_{Si}} \quad (8)$$

To achieve equal channel field conditions, the same K_{TOX} value under the given conditions must be established. By Cramer's rule, the K_{TOX} parameter can be solved through determinant calculation.

$$K_{TOX} = \frac{\det[A_{b(\Delta Q_i, \Delta V_{th})}]}{\det A} \quad (9)$$

where notation $A_{b(\Delta Q_i, \Delta V_{th})}$ stands for 1st column of matrix A replaced with vector $b(\Delta Q_i, \Delta V_{th})$.

Therefore, ΔV_{th} can be solved easily by using the property of multilinearity.

$$\det[A_{b(0, \Delta V_{th})}] = -\Delta V_{th} = \det[A_{b(\Delta Q_i, 0)}] \quad (10)$$

$$\therefore \Delta V_{th} = -\det[A_{b(\Delta Q_i, 0)}] \quad (11)$$

The threshold voltage difference can be calculated with single determinant computation with initially programmed trapped charge distribution is given as Fig. 2(b).

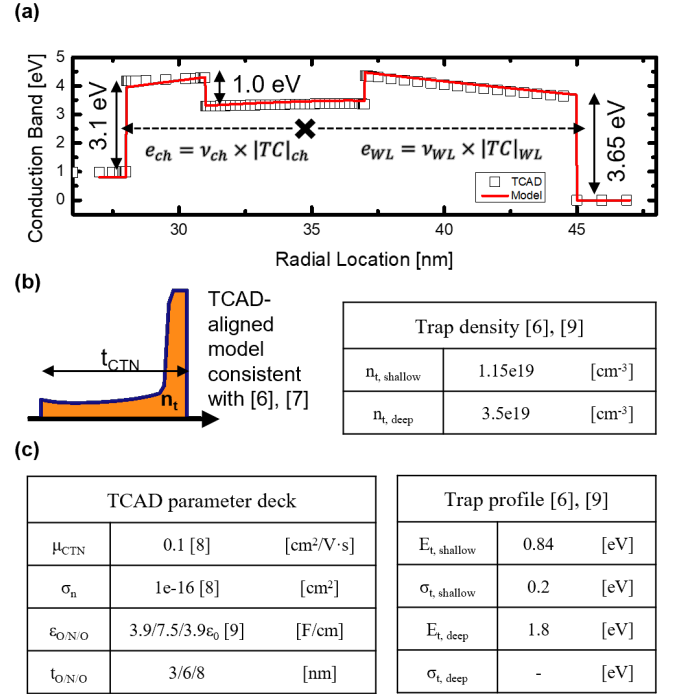


Fig. 2. (a) The band diagram with barrier notations where detrapping of trapped electron occurs, (b) the TCAD-aligned modeled radial trapped-charge profile [6], [7] used as the initial condition of the time-step-free model, and (c) TCAD parameter deck and used trap profile within the simulation [6]–[9].

C. Time-step-free Detrapping Modeling

With the given initial conditions, an autonomous equation is established for each cell, divided by radial position (r) and trap depth (E_t).

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{ch} + e_{WL}) n_t(r, E_t, t) \quad (12)$$

Then, the detrapping relation can be expressed as the following closed-form solution of the autonomous ODE for each radial/energy discretized element.

$$n_t(r, E_t, t) = n_{t0}(r, E_t) \exp(-(e_{ch} + e_{WL})t) \quad (13)$$

The emission rate for trapped electron can be computed as following equations based on a previous study [4].

$$e = \nu \times |TC| \quad (14)$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m_0 \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}} \quad (15)$$

$$|TC| = \prod \exp\left(-\frac{4\sqrt{2m^*} \Phi_H^{1.5} - \Phi_L^{1.5}}{3\hbar q F}\right) \quad (16)$$

for both channel and wordline directions as demonstrated in Fig. 2(a).

This autonomous detrapping equation (13) is the key relation that enables a simulation to directly compute the result at a specific time once initially programmed trapped charge distribution is specified. Therefore, for a fixed spatial/energy discretization and a fixed number of evaluated retention times, the computational cost is governed by the number of trap bins rather than by the final physical retention time.

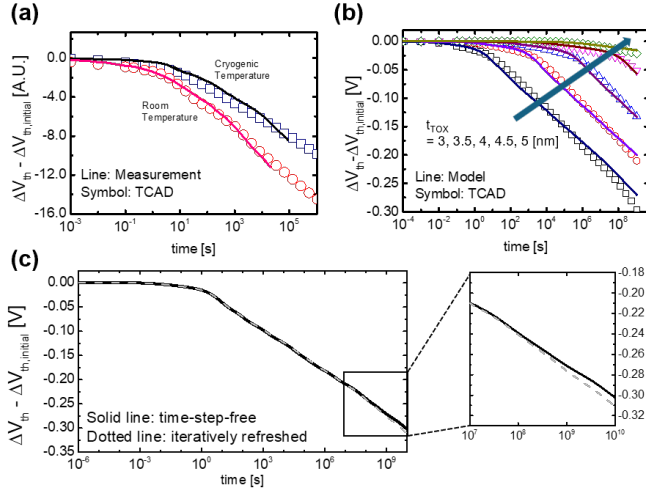


Fig. 3. (a) The TCAD simulation calibration with the measurement from previous study [2]. (b) A detail detrapping curve variation over time with t_{tox} from 3 nm to 5 nm. (c) Comparison between time-step-free simulation and the iterative refresh test for assessing the fixed emission rate approximation.

TABLE I
COMPARISON BETWEEN A TCAD AND TIME-STEP-FREE MODEL

Property	TCAD	Time-step-free model
Temporal treatment	sequential time stepping	closed-form evaluation of eq (13)
Dependence on final physical time	Runtime increases with the final retention time	Runtime independent of final physical time
Run Time (s) (average of Fig. 3(b))	>25000	<4

III. SIMULATION AND RESULT

According to [2], cryogenic conditions enable the detrapping component to be separated from thermally activated retention-loss components. However, direct TCAD simulation at approximately 4.2 K did not converge reliably in our setup. Therefore, two convergence tendencies were compared in the extreme condition: one obtained by lowering the temperature from 300 K to the lowest convergent temperature of 83 K, and the other obtained by reducing the electron capture cross section from 10^{-16} to 10^{-31} cm². As σ_n decreased, the retention transient approached a limiting curve whose time dependence was consistent with the temperature lowering result. Both approaches showed a consistent retention-time trend in the extreme condition, as reflected in the calibrated TCAD curve in Fig. 3(a), indicating that the suppression of thermally activated transitions can reproduce the dominant cryogenic detrapping behavior. Therefore, to emulate cryogenic conditions during retention operation within the TCAD framework, the capture cross section was adjusted to an extremely low value ($\sigma_n \rightarrow 0$) only during retention. This manipulation suppresses thermal electron transitions between trapped and conduction states within the CTN layer, while maintaining physical consistency in other mechanisms [10]. Thus, every TCAD simulation result shown in Fig. 3 should be interpreted as a calibrated cryogenic-emulation benchmark

rather than a direct 4.2 K TCAD simulation.

The TCAD benchmark was built using Synopsys Sentaurus TCAD. The retention deck includes thermionic emission, Poole-Frenkel emission, and nonlocal tunneling, providing a TCAD reference with thermal and tunneling detrapping mechanisms [10]. The present implementation focuses on isolated radial detrapping under cryogenic-emulation conditions, without separately resolving explicit lateral charge migration. The TCAD deck shown in Fig. 2(c) was calibrated by matching the retention time scale of the measurement in previous study [2]. Since ΔV_{th} is reported in arbitrary units, the calibration was done by matching time scale only. Thus, the present validation is limited to the calibrated cryogenic emulation TCAD benchmark and does not establish the absolute experimental cryogenic ΔV_{th} magnitude or full device physics at 4.2 K. Program operation was performed by single 1 μ s pulse and calibrated to obtain $\Delta V_{th} = 1.8$ V. To properly focus on the detrapping effect from the CTN layer, bandgap-engineered processes were neglected and program operation was performed with fresh cell.

Convergence tests over 10-140 trap energy bins and 5-50 radial CTN slices showed negligible changes beyond 100 bins and 30 slices, which were therefore used in the model calculation. The variation of the t_{tox} showed the most dominance in detrapping time transient curve. In Fig. 3(b), t_{tox} from 3 nm to 5 nm for both TCAD simulation and the model were performed, showing good agreement with each other. To verify the time-step-free formulation, an iterative refresh test was performed as shown in Fig. 3(c). In the proposed closed-form calculation, e_{ch} and e_{WL} are evaluated from the initially programmed state and kept fixed. In the refresh calculation, the trapped-charge distribution is sequentially updated after each retention interval. Both approaches show nearly identical ΔV_{th} behavior over the entire retention range, with only about 3% difference from time exceeding 10^9 s. Therefore, the time-step-free simulation was chosen as the main method of this study since the iterative refresh test simply multiplies the simulation time by the total number of iterations.

Table I shows the comparison between TCAD and the model. The table shows how much computation time and resource efficiency could be improved through the time-step-free model. Since the model's computation time is independent of sequential time stepping, the proposed method can be adopted to both short- and long-term retention prediction for arbitrary target retention times.

IV. CONCLUSION

This study presents a time-step-free physics-based modeling for retention loss due to detrapping of trapped electrons. The novelty of time-step-free modeling is demonstrated through significant computational optimization, achieving runtime independent of final physical time. Additionally, the proposed model reproduced the calibrated cryogenic-emulation TCAD benchmark within an extremely short runtime.

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